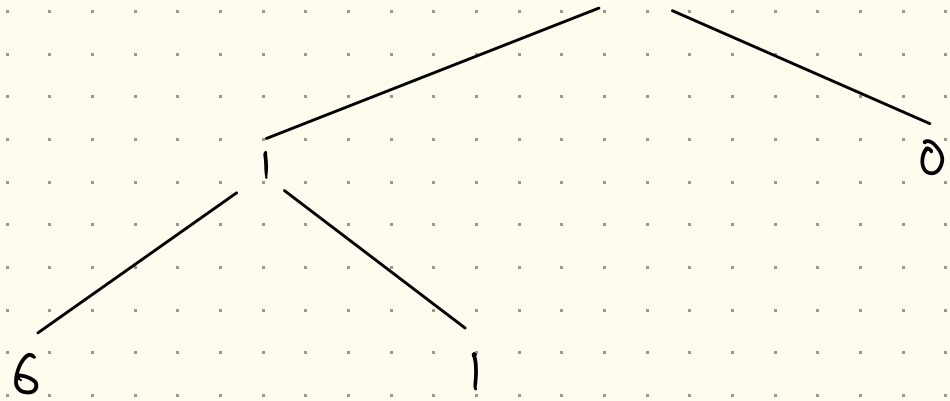
416. Partition Equal Subset Sum

1 5 11 5



⓪ Sum the array

odd -> return False

① See if we can make sum / 2 by summing array elements

BF: TC: 2^n
SC: $O(n)$

BF_CACHED: TC: $O(n \cdot sum(n))$ dfs(i, target)
SC: $O(n \cdot sum(n))$

Recurrence Relation:

1 5 11 5 Given i = 0, if we know all sums that can make in nums[i + 1:]

```
for t in dp:
    if t == target or t == target - nums[i]
        return True
```

Refactor into bottom up dp:

1 5 11 5

init 0

i = 3 -> 5, 0

2 -> ~~16~~, 5, 11, 0